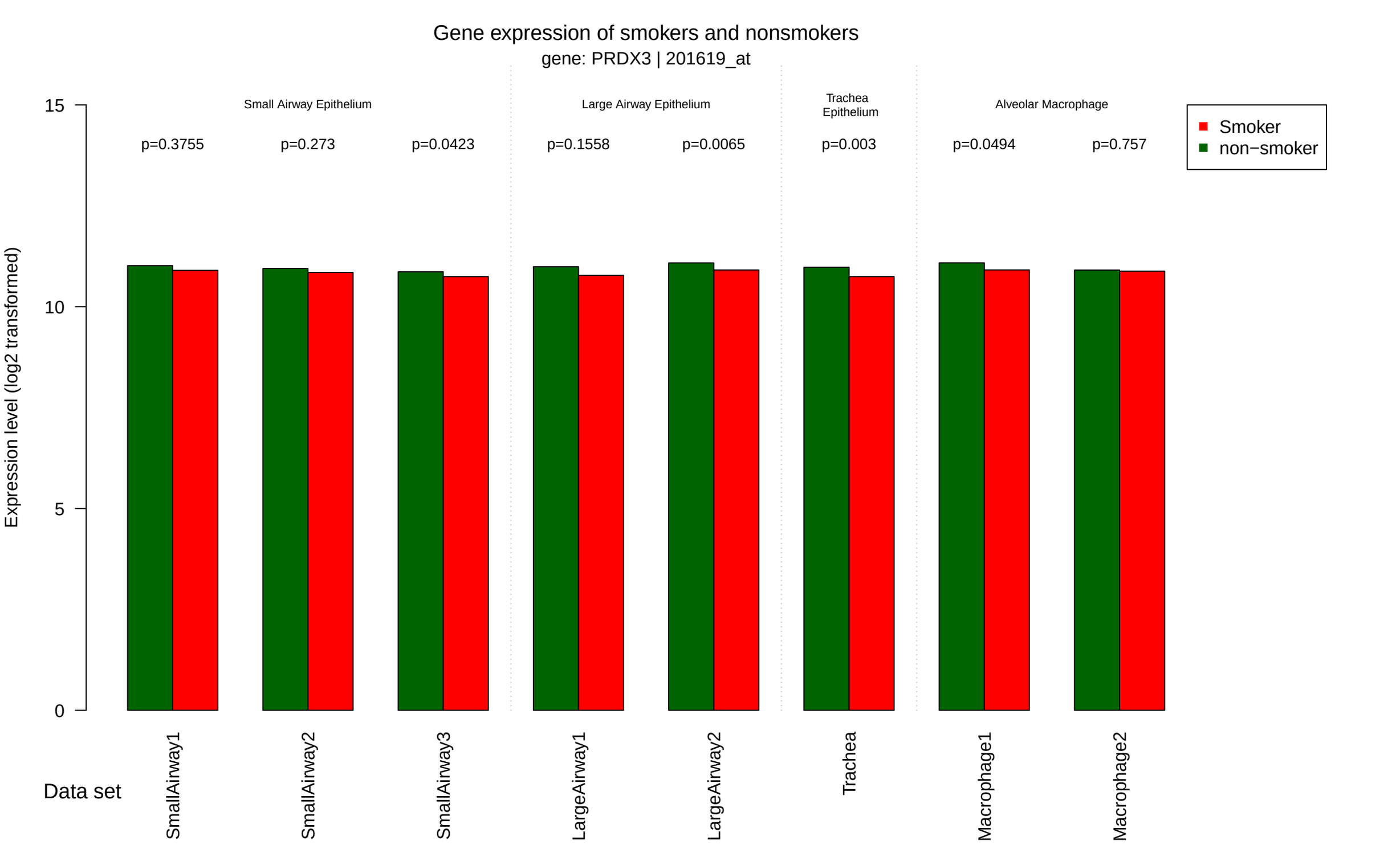
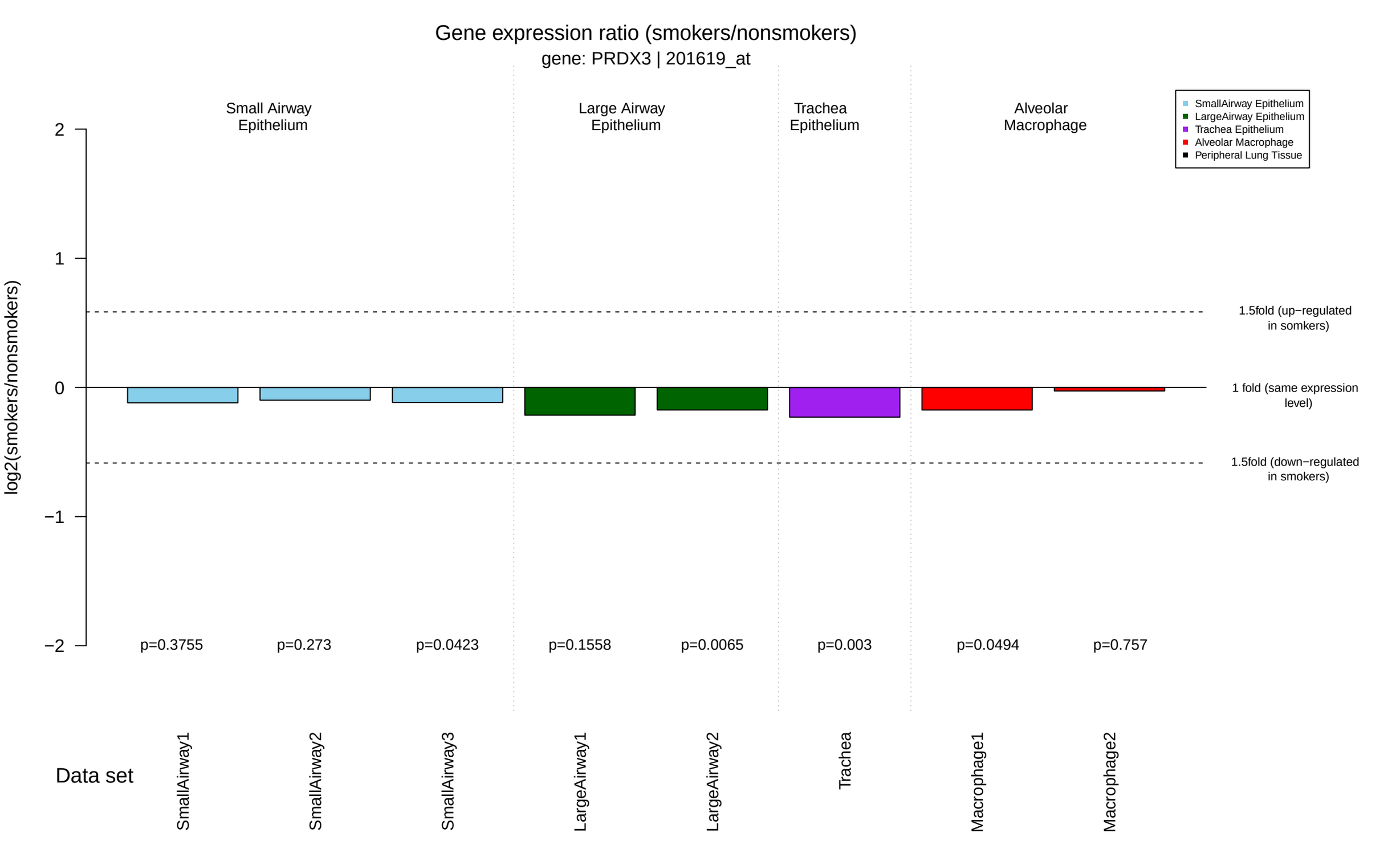
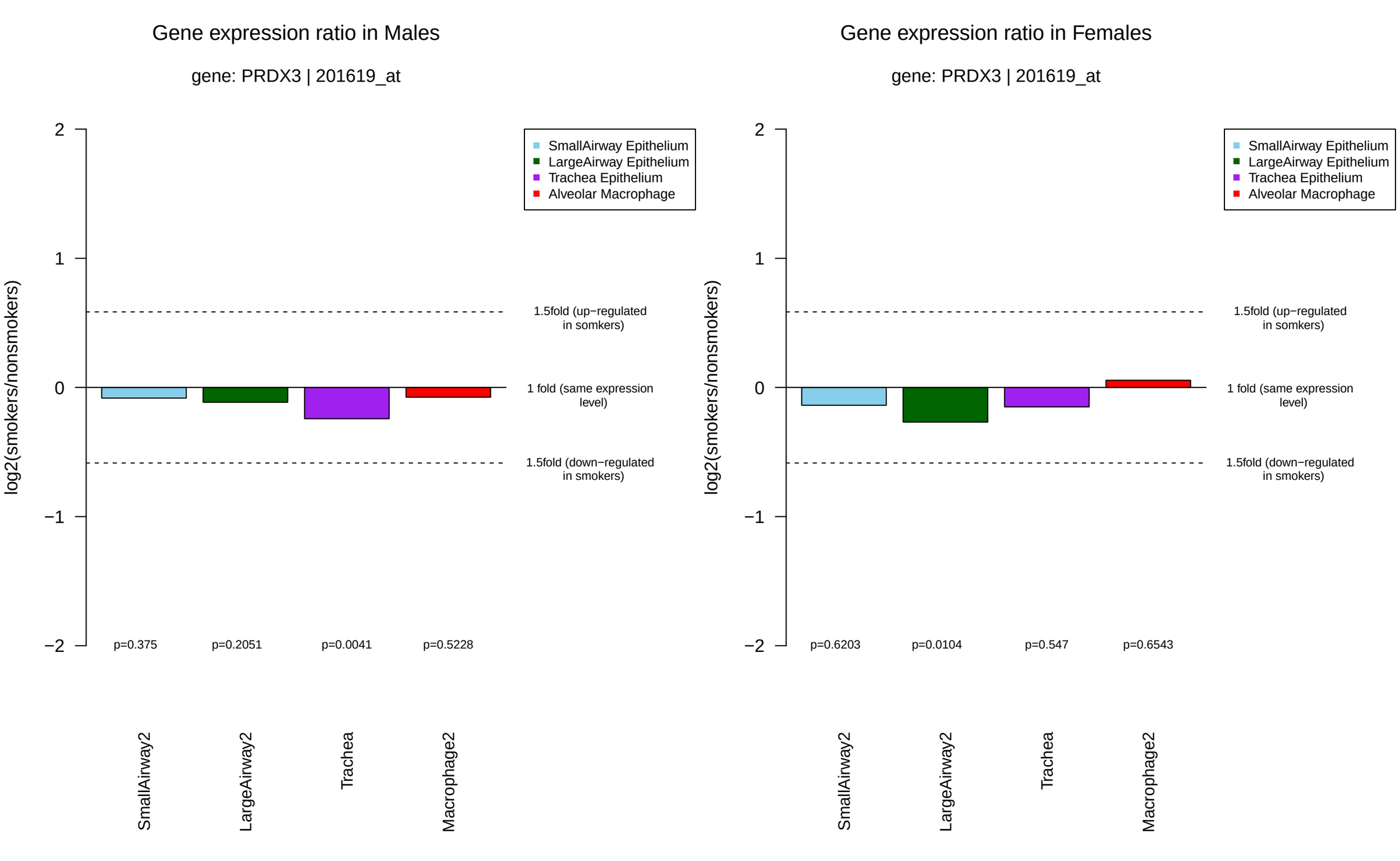
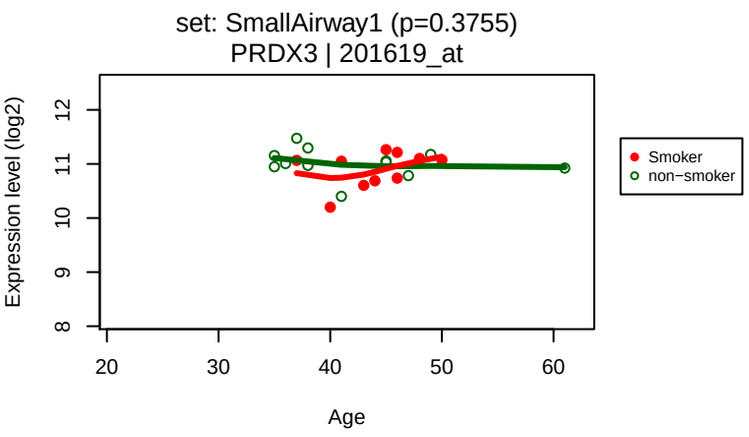
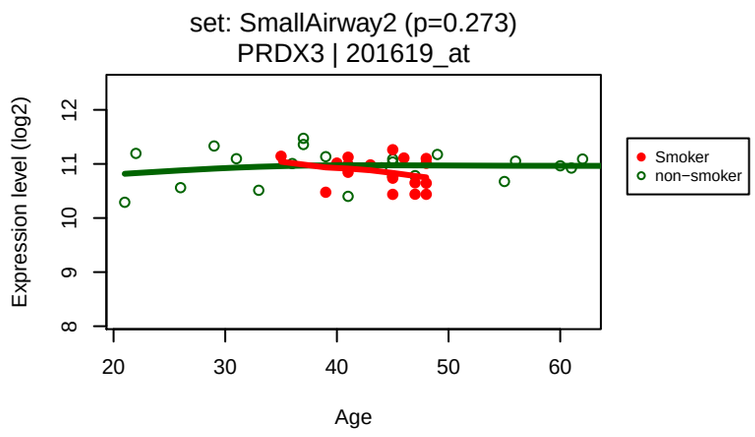
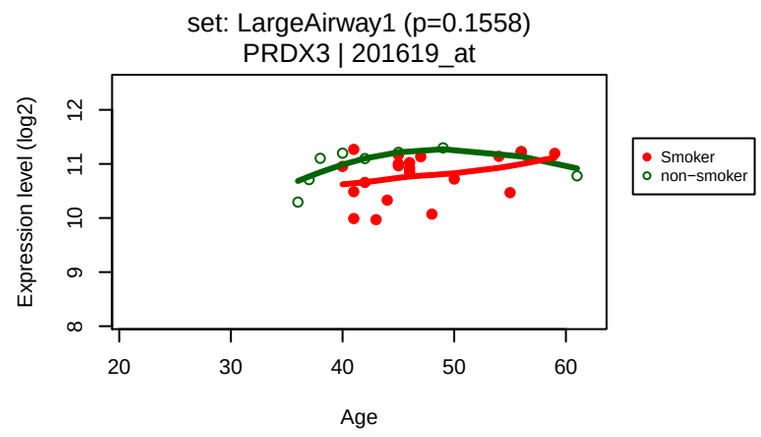

Gene expression of smokers and nonsmokers
gene: PRDX3 | 201619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3755
p=0.273
p=0.0423
p=0.1558
p=0.0065
p=0.003
p=0.0494
p=0.757
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

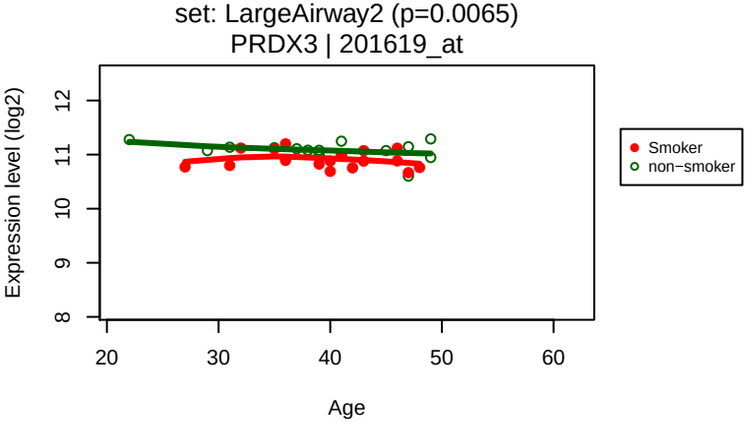

Gene expression ratio (smokers/nonsmokers)
gene: PRDX3 | 201619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3755
p=0.273
p=0.0423
p=0.1558
p=0.0065
p=0.003
p=0.0494
p=0.757
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

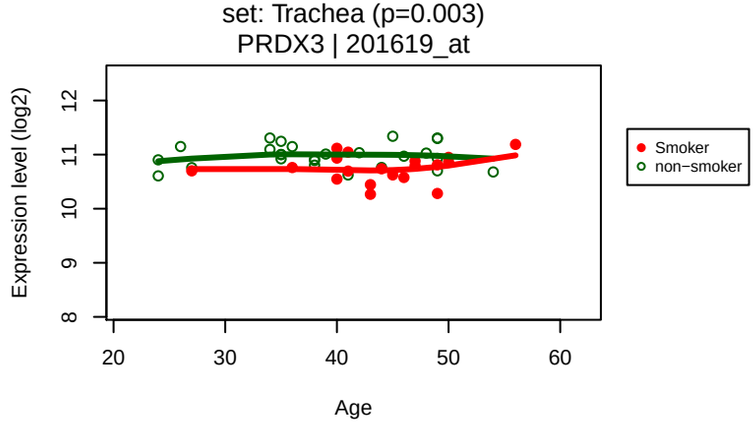

Gene expression ratio in Males
gene: PRDX3 | 201619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.375
p=0.2051
p=0.0041
p=0.5228
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRDX3 | 201619_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6203
p=0.0104
p=0.547
p=0.6543
SmallAirway2
LargeAirway2
Trachea
Macrophage2

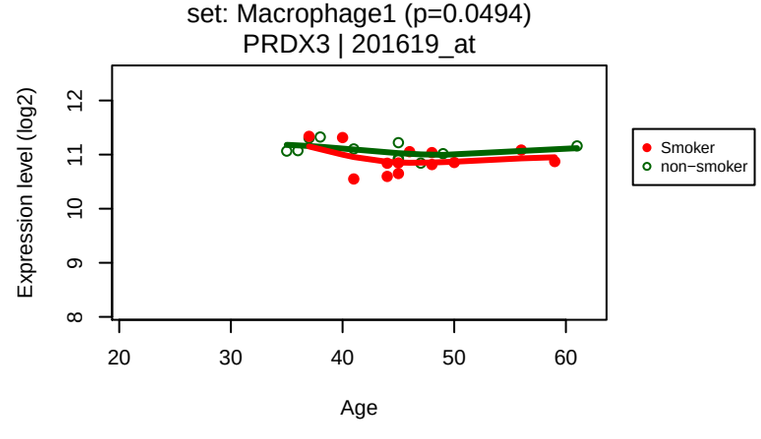

set: SmallAirway1 (p=0.3755)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker

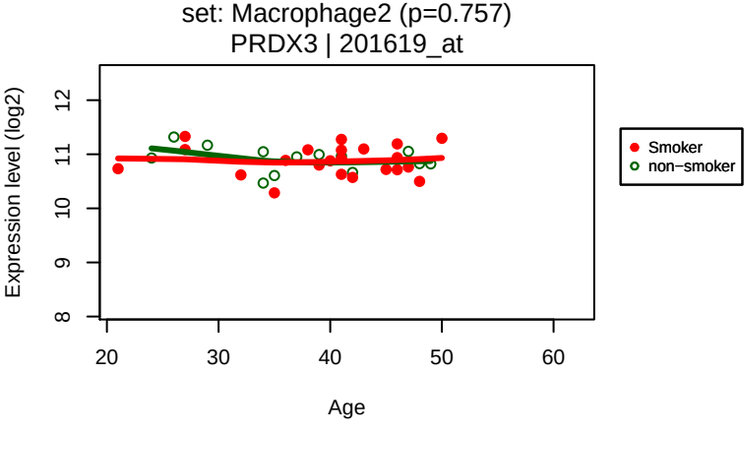

set: SmallAirway2 (p=0.273)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker

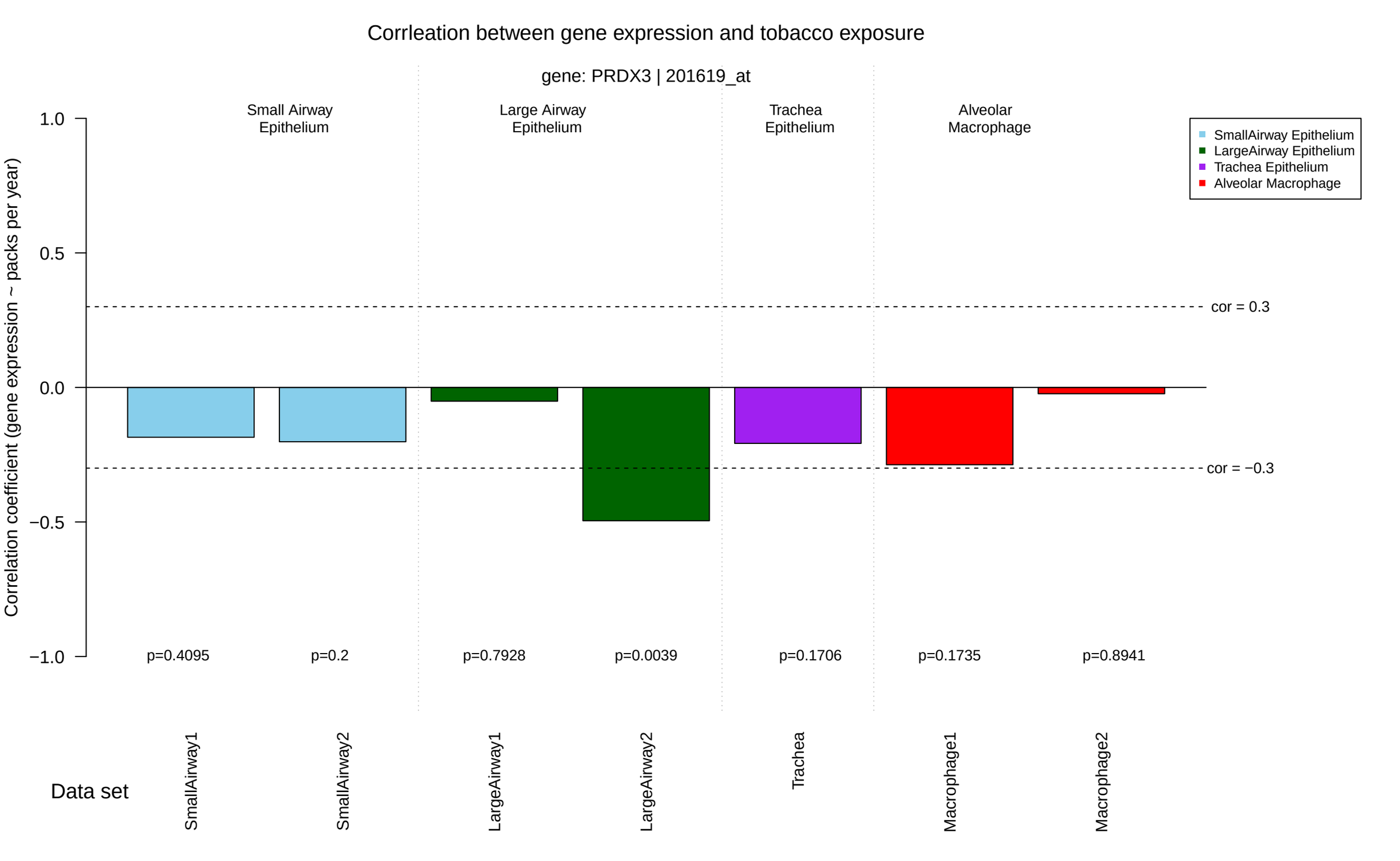

set: LargeAirway1 (p=0.1558)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0065)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.003)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0494)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.757)
PRDX3 | 201619_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRDX3 | 201619_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4095
p=0.2
p=0.7928
p=0.0039
p=0.1706
p=0.1735
p=0.8941
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2